

Enrolment No: _____ Name of Student: _____
 Department/ School: _____

END-TERM MAKEUP EXAMINATION, EVEN SEMESTER 2024

COURSE CODE: CSET206

COURSE NAME: Design and Analysis of Algorithms

PROGRAM: BTECH

MAX. DURATION: 2 HRS

TOTAL MARKS:35

Mapping of Questions to Course and Program Outcomes						
	Section A	Section B				
Q.No.	A1	B1	B2	B3	B4	B5
CO	1	2	3	1	2	3
PO	1-5,9-12	1-5,9-12	1-2,4-5,9-12	1-5,9-12	1-5,9-12	1-2,4-5,9-12
BTL	1	3	3	3	3	3

GENERAL INSTRUCTIONS: -

1. Do not write anything on the question paper except **name, enrolment number** and **department/school**.
2. Carrying mobile phone, smart watch and any other non-permissible materials in the examination hall is an act of **UFM**.

COURSE INSTRUCTIONS: -

1. Attempt **ALL** the questions, **ALL** questions are mandatory.
2. Attempt all **FIVE** questions of **A1** in **Section A**.
3. **Section A** carries **10 marks** and **Section B** carries **25 marks**.
4. Any data you feel missing suitably be assumed and stated clearly.

SECTION A

(5Qx2M = 10 Marks)

A1) Answer all FIVE questions

- I. Identify the recurrence relation $T(n)$ which represents the count of all possible distinct binary strings of length n such that there are no consecutive 1's. also compute the value of $T(5)$.
- II. Suppose Algorithms A and B spend exactly $T_A(n) = C_A n$ and $T_B(n) = C_B n^2$ microseconds, respectively, for a problem of size n . Find the best algorithm for processing $n = 2^{20}$ data items if

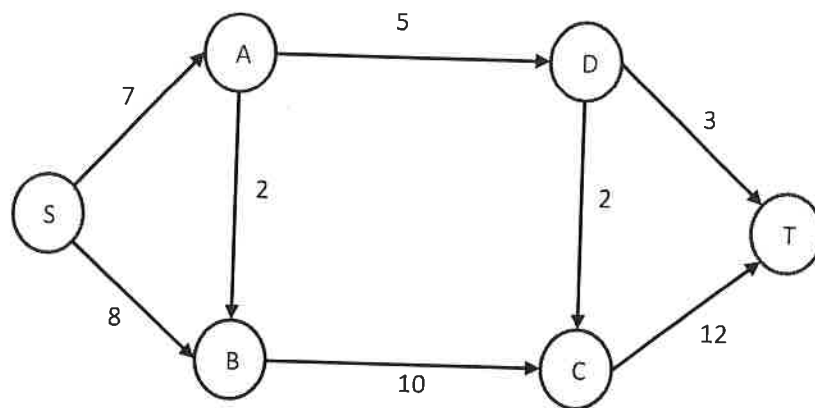
algorithm A spends 10 microseconds to process 1024 items and algorithm B spends only 1 microsecond to process 1024 items (Here C_A and C_B are constants).

- III. Identify whether the following statement is True or False with justification
"Lower bound of a sorting algorithm is $\Omega(n \log_2 n)$ in the worst case to sort n elements."
- IV. Compare Prim's algorithm and Kruskal's algorithm for computing MST.
- V. Identify whether the following statement is True or False with justification
"If $SAT \leq_P A$, then A is NP-complete. Here, A is a computational problem."

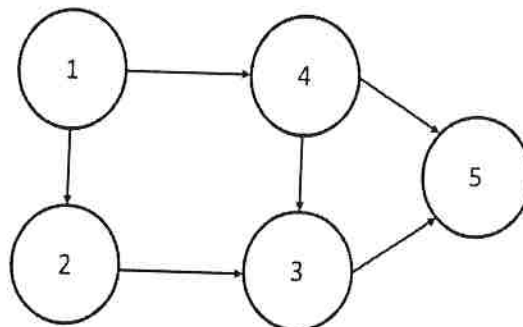
SECTION B

(5Qx5M =25 Marks)

- B1)** Apply the Ford-Fulkerson algorithm step by step to find the maximum flow in the following flow network. (5 Marks)

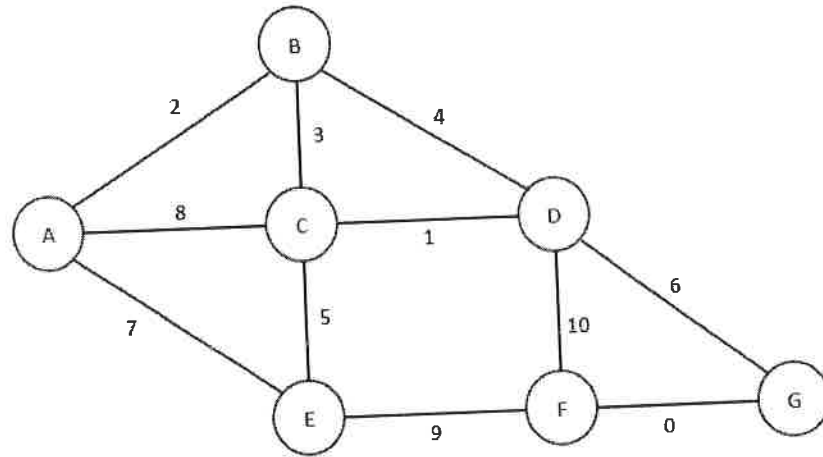


- B2)** I. Explain the topological sorting of a DAG. List all topological sorting of the following given graph (1+1=2 Marks)



- II. Analyse the given graph, where vertices depict cities and edges symbolize network cables for communication, considering the cost associated with laying cables (in Rs). Identify the

optimal network configuration to connect all neighbourhoods, minimizing the overall cost. Identify the chosen edges and compute the total cost incurred for laying cables in this optimal network. **(3 Marks)**

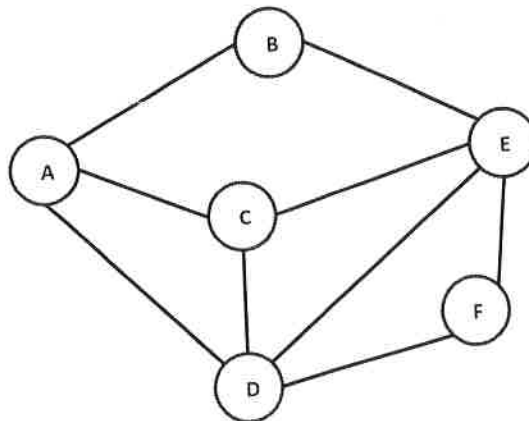


B3) I. Explain P, NP, and NP-Complete class by giving two examples each **(3 Marks)**

II. Analyse the structure of Quicksort, considering splits at each level in the proportion of $1 - \alpha$ to α , where $0 < \alpha \leq 1/2$ is a constant. Compute the minimum depth of a leaf in the recursion tree. **(2 Marks)**

B4) Apply dynamic programming to efficiently multiply all matrices in the provided chain, consisting of matrices A_1, A_2, A_3 , and A_4 with dimensions specified as $p_0=5, p_1=4, p_2=6, p_3=2$, and $p_4=7$. Calculate $m[1,4]$ and determine the optimal order for multiplying all four matrices. **(4+1=5 Marks)**

B5) Explain backtracking with an example, apply backtracking approach to find one Hamiltonian cycle in the following graph starting from A. **(2+3=5 Marks)**



-ALL THE BEST-